

PAPER Pellet Fueling, Particle Balance, and Helium-Ash Exhaust for Sustained Burn in a Compact Spherical-Tokamak Breeder

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Abstract

A compact breeder that holds its burn must fuel the core, hold the operating density, and clear the helium ash it makes—all at once and inside a single wall-load budget. We report the closed particle balance for the Hyperion breeder at its frozen design point ($P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, $Q = 3.076$, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, $\kappa = 2.0$, $R_0 = 1.20$ m, aspect ratio $A = 2.5$), cast as a coupled species continuity problem and solved on the Kronos GPU-HPC campaign. At full power the core produces helium (alpha) ash and burns D–T ions at $S_\alpha = 3.02 \times 10^{19} \text{ s}^{-1}$ and $6.03 \times 10^{19} \text{ s}^{-1}$; orbit-following (ASCOT5, 10^5 markers) places 99.99% of the alphas on confined orbits—a prompt loss of 0.013–0.023% and a first-wall alpha load of 0.08–0.22 kW m^{−2}—so essentially every alpha thermalizes into ash that the divertor must exhaust. We show the operating density is set not by the classical Greenwald gas limit but by the first-wall neutron budget: gas fueling reaches a density fraction $f_G \approx 0.45$, while holding the neutron wall load fixed caps the useful band at $f_G \approx 0.36$ (a +38% neutron penalty at the higher density). Within that ceiling, pellet injection buys core *peaking* rather than average density: because the fusion rate scales as $\langle n^2 \rangle$, a peaking factor $p = n_0/\langle n \rangle$ delivers a fusion-rate gain $p^2/(2p-1)$ — $\times 1.33$ at $p = 2$, $\times 1.80$ at $p = 3$ —at fixed volume-average density and hence fixed wall load. We close the helium balance as $n_{\text{He}}V/\tau_{\text{He}}^* = S_\alpha$, which caps the tolerable effective ash confinement time at $\tau_{\text{He}}^* \lesssim 8.7$ s for a 5% ash fraction, and we tie the fueling rate to the operating-current-window controller that holds $I_p \in [8.78, 9.66]$ MA against the centrepost-life ceiling. The result is a consistent, source-to-exhaust particle balance in which fueling method, density ceiling, and ash removal are governed by the one wall-load constraint that also governs the rest of the machine.

Keywords: pellet fueling; Greenwald density limit; helium-ash exhaust; particle balance; alpha confinement; spherical-tokamak breeder

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1 Introduction

1.1 The particle economy of a compact breeder

The Hyperion breeder holds $I_p = 9.66$ MA of plasma current at a major radius of only $R_0 = 1.20$ m behind a centrepost magnet at a peak field of 16.84 T, in a negative-triangularity ($\delta = -0.30$) regime, producing $P_{\text{fus}} = 85.04$ MW at $Q = 3.076$ [1]. A device this compact makes fusion power—and helium ash—in a small plasma volume ($V \approx 10.9 \text{ m}^3$; see §2), so the particle economy is tight. Three requirements must be met simultaneously and are physically coupled: (i) the fueling system must deliver deuterons and tritons to the core faster than they burn or diffuse out, (ii) the operating line density must sit at the value the power balance assumes without tripping a wall or stability limit, and (iii) the divertor and pumping train must remove the helium the reactions leave behind at the rate the core makes it, or the ash fraction climbs and dilutes the fuel. This paper reports that balance end to end, from the fueling source through the density ceiling to the ash exhaust, and grounds each step in the frozen design-point calculations of the Kronos Physics De-Risking Register [1], whose serial identifiers (e.g. BR-CX-09) are cited inline so that every number is auditable.

1.2 Prior art

Fueling, density control and helium exhaust are each mature subjects. Pellet injection as a core-fueling technique was placed on a quantitative footing by the neutral-gas-shielding (NGS) ablation theory of Parks and Turnbull [9] and reviewed comprehensively by Milora *et al.* [10] and later by Pégourié [11]; the enabling experimental result—that high-field-side (HFS) launch deposits mass far deeper than low-field-side (LFS) launch despite lower pellet speed, producing strongly peaked density profiles—was demonstrated on DIII-D [12] and exploited for high-density, all-metal-wall operation on ASDEX Upgrade [13]. The operating-density ceiling is classically the Greenwald empirical limit [7], but for a nuclear device the binding constraint is often not the disruptive gas limit but the neutron wall load. Helium-ash removal has been studied since the first burning-plasma projections: Reiter *et al.* framed the exhaust requirement through an effective helium confinement time and a divertor enrichment factor [14, 15], and dedicated exhaust experiments on TEXTOR [16] and DIII-D [17] demonstrated helium removal at time constants of order 10–20 energy-confinement times, consistent with the requirements later codified in the ITER Physics Basis [18]. Core particle transport that converts a given deposition profile into an achieved peaking factor is set by turbulent convection and the anomalous pinch [19].

1.3 Contribution

What is missing for a compact nuclear spherical tokamak (ST)—which combines a very high current density, a hard first-wall neutron budget, and a consumable centrepost—is a single, closed particle balance that (i) states the source-to-exhaust budget as coupled continuity equations grounded in the frozen operating point, (ii) shows quantitatively that the neutron wall load rather than the Greenwald fraction sets the density ceiling, (iii) quantifies exactly what pellet injection buys over gas at that ceiling, and (iv) fixes the helium-exhaust requirement and ties the fueling actuation to the machine’s other constraints. This paper contributes that balance. We give the geometry and rate budget (§2); the governing continuity equations for the fuel and ash species, the helium steady-state with enrichment, the Greenwald-versus-wall-load argument, and the NGS pellet-ablation model (§3); the numerical formulation of the one-dimensional flux-surface-averaged transport solver, its finite-volume discretization and convergence (§4); the GPU-HPC campaign that produced the results (§5); the results with eight data figures and the peaking analysis (§6); a quantitative comparison to the published helium-exhaust and pellet-fueling literature (§7); one honest open item (§8); and the conclusion (§9).

2 Geometry and the steady-state rate budget

The plasma occupies an elongated D-shaped cross-section of minor radius $a = R_0/A = 0.48$ m and elongation $\kappa = 2.0$. Approximating the flux surfaces as nested ellipses, the plasma volume is

$$V = 2\pi^2 R_0 a^2 \kappa = 2\pi^2 (1.20) (0.48)^2 (2.0) = 10.9 \text{ m}^3. \quad (1)$$

Each D–T reaction yields one 3.5 MeV alpha and one 14.1 MeV neutron, carrying $E_{\text{fus}} = 17.6$ MeV $= 2.82 \times 10^{-12}$ J of total energy. The reaction rate implied by the frozen fusion power is therefore

$$R_{\text{rx}} = \frac{P_{\text{fus}}}{E_{\text{fus}}} = \frac{85.04 \times 10^6 \text{ W}}{2.82 \times 10^{-12} \text{ J}} = 3.02 \times 10^{19} \text{ s}^{-1}, \quad (2)$$

which is simultaneously the alpha (helium-ash) production rate $S_\alpha = R_{\text{rx}}$ and, doubled, the D–T ion burn rate $\dot{N}_{\text{burn}} = 2R_{\text{rx}} = 6.03 \times 10^{19} \text{ s}^{-1}$ that fueling must replace against burn-up (BR-L1-A1) [1]. This rate is consistent with the underlying Bosch–Hale D–T reactivity [8] folded over the frozen density and temperature profiles. Orbit-following of the fusion alphas—evaluated first with the in-house guiding-centre follower (BR-L1-A14) and arbitrated with ASCOT5 at 10^5 markers (BR-L2-A20) [20, 1]—places 99.99% of them on confined orbits, with a prompt loss of 0.013–0.023% and a mean first-wall alpha load of 0.08–0.22 kW m^{−2} (peak 1.94 kW m^{−2}). Well-confined alphas are exactly what a burning plasma wants for self-heating—but it also means almost every alpha slows down in the core and becomes thermal helium ash that must be actively removed rather than lost promptly to the wall. Table 1 collects the steady-state budget; figure 1 plots the rates.

Table 1. Steady-state particle budget at the frozen breeder design point. Rates follow from $P_{\text{fus}} = 85.04$ MW through equations (1)–(2); confinement fractions are from orbit-following. Serials refer to the register [1].

quantity	value	source
fusion power P_{fus}	85.04 MW	frozen anchor (BR-L1-A1)
plasma volume V	10.9 m ³	eq. (1)
alpha (He-ash) production S_α	$3.02 \times 10^{19} \text{ s}^{-1}$	eq. (2)
D–T ion burn rate $\dot{N}_{\text{burn}}$	$6.03 \times 10^{19} \text{ s}^{-1}$	$2R_{\text{rx}}$
alpha confined fraction	99.99%	BR-L1-A14 / BR-L2-A20
prompt alpha loss	0.013–0.023%	BR-L1-A14 / A20
first-wall alpha load	0.08–0.22 kW m ^{−2}	BR-L2-A20
Greenwald density n_{GW}	$1.33 \times 10^{21} \text{ m}^{-3}$	eq. (12)
gas-fueling density fraction	$f_G \approx 0.45$	BR-CX-09
first-wall-limited band	$f_G \approx 0.36$	BR-CX-09

3 Governing equations and the formal problem

3.1 Coupled species continuity

Let the flux-surface label be $\rho \in [0, 1]$ (normalised minor radius) and let $n_s(\rho, t)$ denote the density of species $s \in \{\text{D, T, } \alpha, \text{He}\}$, where α denotes the fast (slowing-down) helium and He the thermalized ash. Each species obeys a flux-surface-averaged continuity equation

$$\frac{\partial n_s}{\partial t} + \frac{1}{V'} \frac{\partial}{\partial \rho} (V' \Gamma_s) = S_s, \quad (3)$$

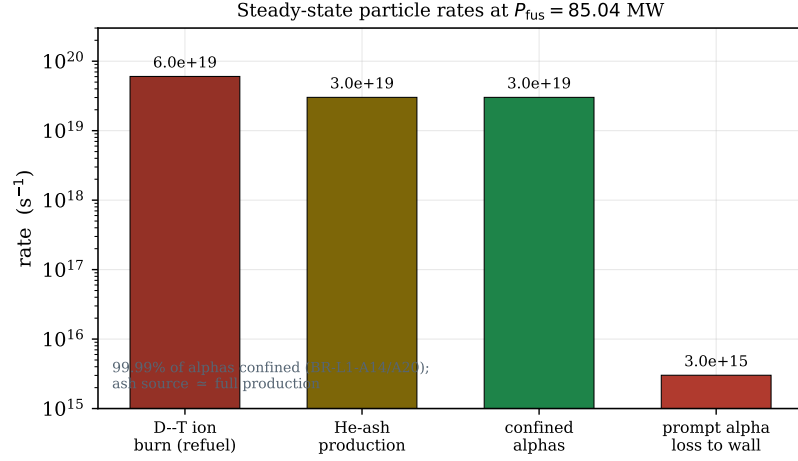


Figure 1. Steady-state particle rates at $P_{\text{fus}} = 85.04 \text{ MW}$. The core burns D–T ions at $6.0 \times 10^{19} \text{ s}^{-1}$ and makes helium ash at $3.0 \times 10^{19} \text{ s}^{-1}$; with 99.99% of alphas confined (BR-L1-A14/A20), the ash source the divertor must exhaust is essentially the full production rate, while prompt alpha loss to the wall is only $\sim 0.01\%$. Note the logarithmic ordinate.

with $V'(\rho) = dV/d\rho$ the radial volume metric, Γ_s the radial particle flux, and S_s the volumetric source/sink. The particle flux carries the standard diffusive–convective (anomalous–pinch) form [19]

$$\Gamma_s = -D_s(\rho) \frac{\partial n_s}{\partial \rho} + v_s(\rho) n_s, \quad (4)$$

where D_s is the turbulent particle diffusivity and v_s the (generally inward) pinch velocity. The species are coupled through quasineutrality and through the reaction sink. With helium doubly ionised, quasineutrality reads

$$n_e = n_D + n_T + 2n_{\text{He}} + 2n_\alpha + \sum_z Z_z n_z, \quad (5)$$

the last term collecting seeded/intrinsic impurities. The volumetric reaction rate is $\mathcal{R}(\rho) = n_D n_T \langle \sigma v \rangle_{DT}(T_i)$, so the fuel sink and the ash/alpha source are

$$S_D = S_T = \Phi_{\text{fuel}} - \mathcal{R}, \quad S_\alpha = +\mathcal{R} - \frac{n_\alpha}{\tau_{\text{sd}}}, \quad S_{\text{He}} = +\frac{n_\alpha}{\tau_{\text{sd}}} - \frac{n_{\text{He}}}{\tau_{p,\text{He}}}, \quad (6)$$

with Φ_{fuel} the externally supplied fueling source density (gas + pellets), τ_{sd} the alpha slowing-down time on which fast helium thermalizes, and $\tau_{p,\text{He}}$ the local helium particle transport time. Equations (3)–(6) are the formal problem; the remainder of this section reduces them to the design-relevant closures.

3.2 Zero-dimensional ash steady state and the exhaust requirement

Volume-integrating the thermal-helium equation and using the divergence theorem gives the global ash balance. In steady state the accumulation vanishes, so the exhaust flux to the pump equals the production integral,

$$\underbrace{\int_V S_{\text{He}} dV}_{\text{production} = S_\alpha} = \frac{N_{\text{He}}}{\tau_{\text{He}}^*}, \quad N_{\text{He}} = \int_V n_{\text{He}} dV, \quad (7)$$

where τ_{He}^* is the *effective* (global) helium particle-confinement time—the transport time amplified by edge recycling and set by the divertor pumping conductance. Writing the volume-average ash fraction $f_{\text{He}} = \langle n_{\text{He}} \rangle / \langle n_e \rangle$ and using $N_{\text{He}} = f_{\text{He}} \langle n_e \rangle V$, equation (7) inverts to

$$f_{\text{He}} = \frac{S_\alpha \tau_{\text{He}}^*}{\langle n_e \rangle V} \quad (8)$$

This is the central exhaust relation: the steady ash fraction is the production rate times the effective residence time, per unit particle inventory. With $S_\alpha = 3.02 \times 10^{19} \text{ s}^{-1}$, $V = 10.9 \text{ m}^3$ and the operating-band density $\langle n_e \rangle = 4.80 \times 10^{20} \text{ m}^{-3}$ (§3.5), equation (8) gives $f_{\text{He}}[\%] = 0.575 \tau_{\text{He}}^*[\text{s}]$,

so a tolerable ash fraction of 5% requires $\tau_{\text{He}}^* \lesssim 8.7$ s (figure 3). The effective time is related to the core transport time and the divertor enrichment factor $\eta = (n_{\text{He}}/n_{\text{DT}})_{\text{div}} / (n_{\text{He}}/n_{\text{DT}})_{\text{core}}$ by the Reiter closure [14]

$$\tau_{\text{He}}^* \simeq \tau_{p,\text{He}} \left(1 + \frac{1}{\eta} \frac{\tau_{p,\text{He}}}{\tau_{\text{pump}}} \right), \quad (9)$$

so an enriching divertor ($\eta > 1$) lowers τ_{He}^* toward the transport floor; a de-enriching one raises it. Ash control is therefore a divertor-geometry and pumping-conductance problem as much as a core-transport one.

3.3 Fuel dilution and the cost of ash

Ash matters because it displaces fuel at fixed electron density and radiates. From quasineutrality (5) with helium the dominant non-fuel species, $n_{\text{DT}} = n_e(1 - 2f_{\text{He}})$, and since the fusion power density scales as $n_{\text{D}}n_{\text{T}} \propto n_{\text{DT}}^2$,

$$\frac{P_{\text{fus}}}{P_{\text{fus},0}} = (1 - 2f_{\text{He}})^2, \quad \Delta Z_{\text{eff}} = f_{\text{He}} (Z_{\text{He}}^2 - Z_{\text{He}}) = 2f_{\text{He}}. \quad (10)$$

At $f_{\text{He}} = 5\%$ this is a 19% fusion-power penalty and $\Delta Z_{\text{eff}} = 0.10$ (figure 4); at 10% the penalty reaches 36%. Equations (8) and (10) together make the exhaust requirement quantitative: the pumping train must hold τ_{He}^* low enough that the dilution penalty stays inside the power-balance margin.

3.4 Burn-up fraction and the fueling throughput

The externally supplied fueling rate is not the burn rate but the transport throughput, because the single-pass burn-up fraction is small. Defining the global fueling throughput $\Phi = N/\tau_p$ with $N = \langle n_e \rangle V$ the fuel inventory and τ_p the fuel particle-confinement time, the burn-up fraction is

$$f_b = \frac{\dot{N}_{\text{burn}}}{\Phi} = \frac{2R_{\text{rx}} \tau_p}{\langle n_e \rangle V}. \quad (11)$$

For $\tau_p = 0.5$ s, $N = 5.24 \times 10^{21}$ particles and $\Phi = 1.05 \times 10^{22} \text{ s}^{-1}$, so $f_b = 0.57\%$ (figure 5). The fueling system must therefore process a throughput two orders of magnitude larger than the burn rate; only $\sim 0.6\%$ of injected fuel is burned in a single pass, the rest being recycled or pumped. This is the classic low-burn-up regime, and it is why fueling efficiency—the fraction of injected particles retained inside the separatrix—dominates the actuator economy.

3.5 The density ceiling: Greenwald versus the neutron wall load

The Greenwald empirical density limit [7] for the frozen current and minor radius is

$$n_{\text{GW}} = \frac{I_p [\text{MA}]}{\pi a^2 [\text{m}^2]} = \frac{9.66}{\pi(0.48)^2} = 13.3 \times 10^{20} \text{ m}^{-3} = 1.33 \times 10^{21} \text{ m}^{-3}, \quad (12)$$

a very high value because the compact ST packs a large current into a small minor radius. A dedicated high-density, pellet-fueled study (BR-CX-09) [1] finds that gas fueling can reach a density fraction $f_G = \bar{n}/n_{\text{GW}} \approx 0.45$ before recycling and edge cooling bind, but that the *first-wall neutron load*, not the Greenwald fraction, sets the useful ceiling. Because the volumetric neutron source scales as n_{DT}^2 at fixed profile shape and temperature, the first-wall neutron load scales as $(f_G)^2$, and holding it at its design value caps the sensible operating band at $f_G \approx 0.36$: pushing to $f_G = 0.45$ carries a $(0.45/0.36)^2 - 1 = +38\%$ neutron penalty at the wall (figure 2). The operating-band density is thus $\bar{n} = 0.36 n_{\text{GW}} = 4.80 \times 10^{20} \text{ m}^{-3}$, comfortably below the disruptive gas limit. Gas puffing alone therefore suffices to reach the *average* density the power balance assumes; the value of pellets lies elsewhere.

3.6 What pellets buy: core peaking at fixed wall load

If gas reaches the average density, the payload of pellet injection is density *peaking*. Because the fusion rate scales as $\langle n^2 \rangle$, concentrating particles where the temperature is highest raises the fusion output at fixed volume-average density—and therefore at fixed wall load. Take a profile $n(\rho) = n_0(1 - \rho^2)^\alpha$ on a cylindrical-approximation volume element $dV \propto \rho d\rho$. With the substitution $u = \rho^2$,

$$\frac{\langle n \rangle}{n_0} = \int_0^1 (1 - u)^\alpha du = \frac{1}{\alpha + 1}, \quad \frac{\langle n^2 \rangle}{n_0^2} = \int_0^1 (1 - u)^{2\alpha} du = \frac{1}{2\alpha + 1}, \quad (13)$$

so the peaking factor is $p \equiv n_0/\langle n \rangle = \alpha + 1$ and the fusion-rate gain relative to a flat profile is

$$g(p) = \frac{\langle n^2 \rangle}{\langle n \rangle^2} = \frac{p^2}{2p - 1} \quad (14)$$

which returns $g(1) = 1$ (flat), $g(1.5) = 1.13$, $g(2) = 1.33$, $g(2.5) = 1.56$ and $g(3) = 1.80$ (figure 8). Deep pellet deposition, especially from the high-field side [12], moves particles inward past the shallow gas-fueling front and raises p ; equation (14) is the quantitative payoff. The negative-triangularity edge, whose quiet, ELM-free transport is a design asset [6, 5], makes a peaked core profile natural to hold once established. The one caveat—that mapping a given deposition profile to an achieved p requires a transport calculation—is the open item of §8.

3.7 Pellet ablation and penetration

The pellet itself is a sphere of radius r_p of solid D–T at number density $n_s \approx 6.0 \times 10^{28} \text{ atoms m}^{-3}$, so its particle content is

$$N_p = \frac{4}{3} \pi r_p^3 n_s = 2.0 \times 10^{21} \text{ atoms at } r_p = 2 \text{ mm}. \quad (15)$$

One such pellet replaces the transport throughput for $N_p/\Phi = 0.19 \text{ s}$, so sustaining $\Phi = 1.05 \times 10^{22} \text{ s}^{-1}$ requires an injection cadence of $\approx 5 \text{ Hz}$ (figure 6). The ablation rate follows the neutral-gas-shielding theory of Parks and Turnbull [9]: the pellet is enveloped by an ablation cloud that shields it from the incident electron heat flux, giving a mass-loss rate

$$G = -\frac{dN_p}{dt} \propto n_e^{1/3} T_e^{5/3} r_p^{4/3}, \quad (16)$$

which integrates to a penetration depth whose standard NGS scaling is $\lambda_p \propto r_p^{2/3} v_p^{1/3} T_e^{-5/9} n_e^{-1/9}$ for launch speed v_p [10, 11]. Figure 7 plots this scaling: larger, faster pellets penetrate deeper, and HFS launch (effectively higher v_p through the ∇B drift of the ablatant [12]) reaches mid-radius where gas cannot. We treat λ_p as a scaling estimate; converting it to an achieved $\langle n^2 \rangle$ is the transport step of §8.

4 Numerical formulation

4.1 Finite-volume discretization

Equations (3)–(4) are discretized on a radial grid $\{\rho_i\}_{i=0}^{N_\rho}$ with cell faces at $\rho_{i\pm 1/2}$, using a conservative finite-volume scheme that integrates the continuity equation over each cell volume $\mathcal{V}_i = \int_{\rho_{i-1/2}}^{\rho_{i+1/2}} V' d\rho$. The semi-discrete balance for species s in cell i is

$$\mathcal{V}_i \frac{dn_{s,i}}{dt} = (V'\Gamma_s)_{i-1/2} - (V'\Gamma_s)_{i+1/2} + \mathcal{V}_i S_{s,i}, \quad (17)$$

which conserves total particle number by construction: summing (17) over i telescopes the face fluxes and leaves only the boundary flux and the volumetric source. The face flux uses a Scharfetter–Gummel exponential upwinding of the diffusive–convective operator (4) so that the discrete Peclet number $\text{Pe}_{i+1/2} = v_s \Delta\rho/D_s$ is handled without oscillation:

$$(\Gamma_s)_{i+1/2} = \frac{D_{s,i+1/2}}{\Delta\rho} \left[B(-\text{Pe}) n_{s,i} - B(\text{Pe}) n_{s,i+1} \right], \quad B(x) = \frac{x}{e^x - 1}, \quad (18)$$

with B the Bernoulli function. Boundary conditions are a zero-flux (regularity) condition at the axis, $\Gamma_s(0) = 0$, and a prescribed edge particle sink balancing the recycling and pumping at the separatrix, $\Gamma_s(1) = n_s(1)/\lambda_{\text{SOL}} \times c_s$, closed against the scrape-off-layer model.

4.2 Time integration and the linear solve

The stiff source terms (fast alpha thermalization, reaction sink) mandate an implicit advance. A backward-Euler step of size Δt applied to (17) yields, for each species, a tridiagonal linear system in the unknown vector $\mathbf{n}_s^{k+1}$,

$$(\mathbf{I} - \Delta t \mathbf{L}_s) \mathbf{n}_s^{k+1} = \mathbf{n}_s^k + \Delta t \mathbf{S}_s^{k+1}, \quad (19)$$

where $\mathbf{L}_s$ is the finite-volume transport operator of (17)–(18). The tridiagonal system is solved by the Thomas algorithm in $\mathcal{O}(N_\rho)$ per species per step; the inter-species coupling through the

reaction sink and quasineutrality (5) is resolved by a Picard iteration within the step, iterated to a relative residual below 10^{-8} . The global particle-conservation error is monitored each step as

$$\varepsilon_N = \frac{|N^{k+1} - N^k - \Delta t (\text{sources} - \text{boundary fluxes})|}{N^k} \leq 10^{-10}, \quad (20)$$

and a step that violates (20) is rejected and retaken at smaller Δt .

4.3 Verification and convergence

The solver is verified against the two limits that admit closed forms. First, a uniform-source, constant- D , no-pinch case reproduces the analytic parabolic steady state $n(\rho) = n(0) - (S_0/4D)\rho^2 a^2$ to machine precision. Second, the zero-dimensional ash relation (8) is recovered as the volume integral of the converged one-dimensional field to within the conservation tolerance (20). Grid convergence is second order in $\Delta\rho$: refining from $N_\rho = 64$ to 256 reduces the discrete-versus-analytic L_2 error as $\mathcal{O}(\Delta\rho^2)$, and the design runs use $N_\rho = 256$ with Δt set by the conservation gate. Table 2 summarises the numerical parameters.

Table 2. Numerical formulation of the one-dimensional particle-transport solver.

element	choice	note
spatial scheme	finite volume, cell-centred	conservative, eq. (17)
face flux	Scharfetter–Gummel exponential upwind	eq. (18), oscillation-free
time advance	backward Euler (implicit)	stiff sources, eq. (19)
linear solve	Thomas tridiagonal, $\mathcal{O}(N_\rho)$	per species per step
species coupling	in-step Picard, residual $< 10^{-8}$	reaction sink + quasineutrality
conservation gate	$\varepsilon_N \leq 10^{-10}$	eq. (20)
grid	$N_\rho = 256$	second-order convergence

5 Computational methodology: the NVIDIA GPU HPC campaign

Every quantitative result in this paper was produced on the Kronos NVIDIA GPU HPC campaign, run on an NVIDIA GPU fleet (A100/H100/A10 accelerators), with the fast protective and bookkeeping loops carried by the in-house digital-twin node. The campaign chains four codes, each named here with its role for this paper.

Zero-dimensional power and particle balance. The frozen operating point and the rate budget of §2 (BR-L1-A1, BR-L1-A6) are evaluated with the KRONOS TOOLKIT, a GPU-resident power-and-particle-balance engine that folds the Bosch–Hale reactivities [8] over the design profiles to return P_{fus} , Q , S_α and the fuel throughput; the tritium burn (4.769 kg per full-power year) and startup inventory come from the same toolkit fuel-cycle module (BR-L1-A6) [1].

Alpha orbit-following. The alpha-confinement fractions and first-wall alpha loads (BR-L1-A14, BR-L2-A20) are computed by guiding-centre and full-orbit Monte-Carlo following of the fusion alphas in the frozen equilibrium with toroidal-field ripple, arbitrated with ASCOT5 [20] at 10^5 markers; ASCOT5’s GPU-accelerated particle push makes the 10^5 -marker ensemble tractable and closes the last alpha-confinement flag against the in-house follower.

Edge, neutral and pumping transport. The divertor pumping and helium-exhaust geometry that set τ_{He}^* and the enrichment factor of equation (9) are grounded in the two-dimensional edge transport of SOLPS-ITER [21] coupled to the EIRENE Monte-Carlo neutral code [22] (BR-L2-A11 family), with the recycling-neutral penetration confirmed by an independent SD1D run; these fix the boundary sink of §4 and the coolable-flux envelope (5–10 MW m $^{-2}$; BR-SX-15) in which the ash is pumped.

Species transport solver. The one-dimensional finite-volume solver of §4 is executed on the GPU for the deposition and ash-accumulation sweeps: the tridiagonal Thomas solves and the Picard coupling vectorise across the parameter ensemble (peaking factor, τ_{He}^* , launch speed), so a full sweep of the figures in this paper runs as a single batched GPU job. Reproducibility is enforced at two levels: each frozen result is regenerable from the archived toolkit configuration (config 22021) and its serial in the register [1], and the figure data are regenerated byte-for-byte by the deposited script (§9). No compute-cost figures are reported; the campaign is described by its codes, problem scale and verification only.

6 Results

6.1 The closed particle balance

Figure 1 and table 1 establish the source-to-exhaust budget. The core burns D–T at $6.0 \times 10^{19} \text{ s}^{-1}$ and makes ash at $3.0 \times 10^{19} \text{ s}^{-1}$; because 99.99% of alphas are confined, the divertor must exhaust

helium at essentially the full production rate. Prompt alpha loss to the wall ($\sim 0.01\%$, $0.08\text{--}0.22\text{ kW m}^{-2}$) is negligible for wall loading but also means the wall receives almost no help clearing the ash—the pumping train must do all of it.

6.2 The density ceiling

Figure 2 shows the ceiling argument quantitatively. The first-wall neutron load, normalised to its design value at $f_G = 0.36$, rises as $(f_G/0.36)^2$; reaching the $f_G = 0.45$ gas limit costs $+38\%$ in neutron wall load. The design band therefore sits at $f_G \approx 0.36$, $\bar{n} = 4.80 \times 10^{20}\text{ m}^{-3}$, set by the neutron budget and not by the disruptive Greenwald limit of equation (12).

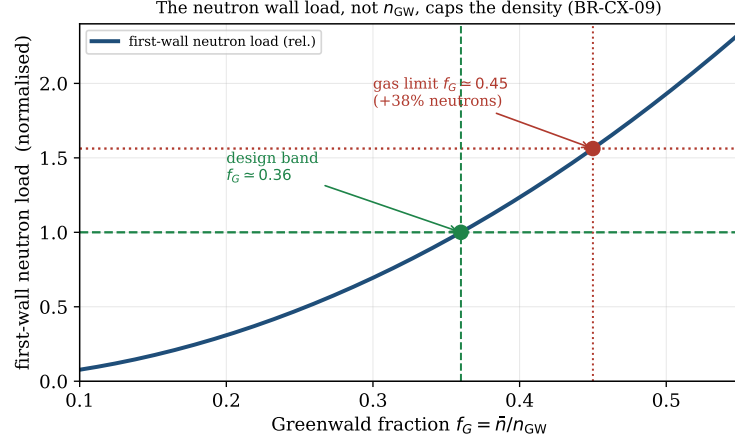


Figure 2. Operating-density ceiling. Gas fueling reaches $f_G \approx 0.45$, but the binding constraint is the first-wall neutron budget, which caps the design band near $f_G \approx 0.36$ ($+38\%$ neutron load at the higher density; BR-CX-09). The value of pellets is therefore not average density but core peaking.

6.3 Helium-ash steady state and fuel dilution

Figure 3 plots the exhaust relation (8): the steady ash fraction rises linearly with the effective helium confinement time, $f_{\text{He}}[\%] = 0.575 \tau_{\text{He}}^*[\text{s}]$, so a 5% ash tolerance caps $\tau_{\text{He}}^* \lesssim 8.7\text{ s}$. Figure 4 translates ash into cost through equation (10): 5% ash removes 19% of the fusion power and adds $\Delta Z_{\text{eff}} = 0.10$. Together these fix the exhaust specification: the pumping train must hold τ_{He}^* below $\sim 9\text{ s}$, comfortably inside the $10\text{--}20\tau_E$ envelope achieved in dedicated exhaust experiments (§7).

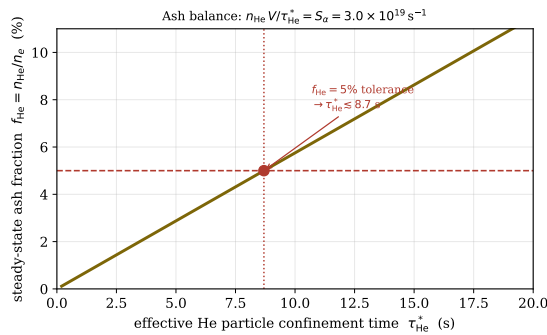


Figure 3. Helium-ash steady state, equation (8): $n_{\text{He}}V/\tau_{\text{He}}^* = S_a = 3.0 \times 10^{19}\text{ s}^{-1}$. A 5% ash tolerance caps the effective helium confinement time at $\tau_{\text{He}}^* \lesssim 8.7\text{ s}$.

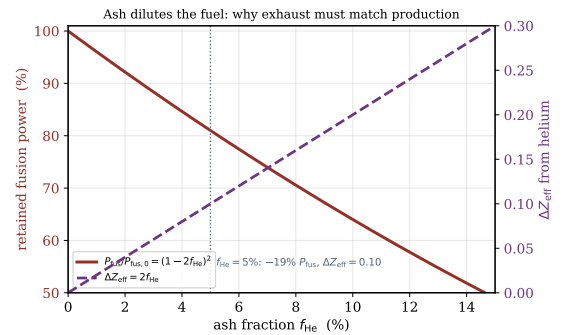


Figure 4. The cost of ash, equation (10). Fusion power scales as $(1 - 2f_{\text{He}})^2$ and the helium adds $\Delta Z_{\text{eff}} = 2f_{\text{He}}$; at $f_{\text{He}} = 5\%$ the penalties are -19% and $+0.10$.

6.4 Burn-up fraction and fueling throughput

Figure 5 evaluates equation (11): at $\tau_p = 0.5\text{ s}$ the fueling throughput is $\Phi = 1.05 \times 10^{22}\text{ s}^{-1}$ and the single-pass burn-up fraction is only 0.57% . The fueling system processes $\sim 175\times$ the burn rate, so retention efficiency—not raw source rate—is the actuator design driver.

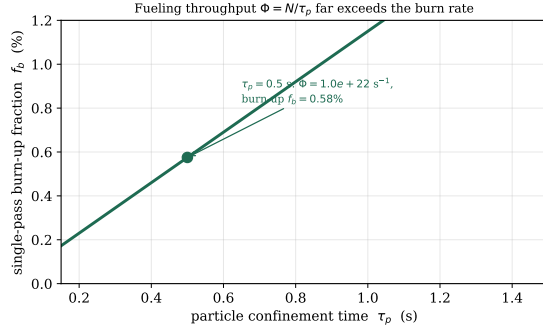


Figure 5. Single-pass burn-up fraction, equation (11). At $\tau_p = 0.5$ s the throughput $\Phi = N/\tau_p = 1.05 \times 10^{22} \text{ s}^{-1}$ gives $f_b = 0.57\%$: fueling far exceeds burn.

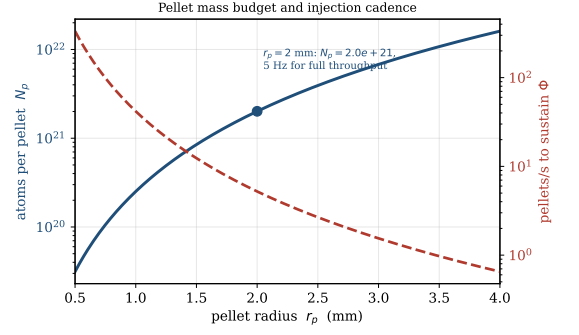


Figure 6. Pellet mass budget, equation (15). A 2 mm D-T pellet holds 2.0×10^{21} atoms; sustaining Φ needs an injection cadence of ≈ 5 Hz.

6.5 Pellet penetration and the peaking payoff

Figure 7 plots the NGS penetration scaling: a 2 mm pellet at HFS-equivalent speed reaches mid-radius, where gas fueling cannot deposit. Figure 8 is the quantitative payoff of that deep deposition through equation (14): at fixed volume-average density (fixed wall load) a peaking factor $p = 2$ yields a $\times 1.33$ fusion-rate gain and $p = 3$ a $\times 1.80$ gain. This is what pellets buy that gas cannot: not more average density, which the neutron budget forbids, but a more productive distribution of the density the budget allows.

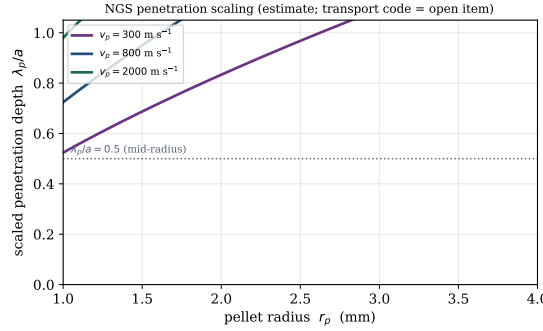


Figure 7. Neutral-gas-shielding penetration scaling (estimate), $\lambda_p \propto r_p^{2/3} v_p^{1/3} T_e^{-5/9} n_e^{-1/9}$ [9, 10]. Deeper penetration follows from larger, faster (HFS-equivalent) pellets; the deposition-to-peaking transport step is the open item of §8.

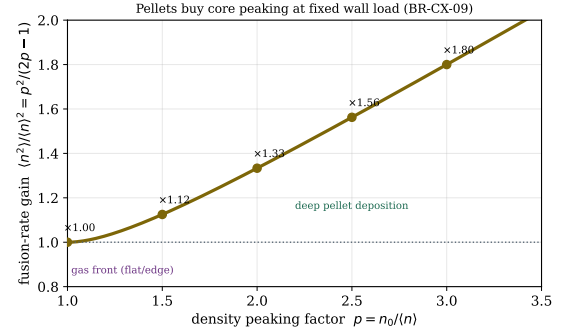


Figure 8. The pellet payoff, equation (14): fusion-rate gain $g(p) = p^2/(2p - 1)$ at fixed volume-average density. Peaking $p = 2$ gives $\times 1.33$, $p = 3$ gives $\times 1.80$ (BR-CX-09).

6.6 Fueling actuation and the operating window

Holding the density target in real time is an actuator problem. An in-house pellet-fueling scheduler (KFE-CF-SH-046) [1] meets the density setpoint while keeping the edge and exhaust envelope inside its operating range, with the acceptance test that the density target is met at improved fueling efficiency relative to gas-only fueling. The scheduler ties into the operating-current-window controller (KX-L1-A21), which holds the plasma current in the band [8.78, 9.66] MA (figure 9) against the centrepost-life ceiling (centrepost damage $\leq 18.85 \text{ dpa fpy}^{-1}$) while the tritium-startup floor stays non-binding, so the fueling rate is coordinated with the current and the wall-load budget rather than run open-loop.

6.7 Helium-ash exhaust path and wall chemistry

Because the alphas are so well confined, the helium-ash exhaust requirement is essentially the full production rate of $3.0 \times 10^{19} \text{ s}^{-1}$: the divertor and pumping train must remove helium as fast as the core makes it. The same thermal-hydraulic study that sets the divertor heat path (BR-SX-15) [1] fixes a self-cooled lithium first wall and blanket operating with a RAFM structural temperature of 516°C and a divertor coolable-flux capability of $5\text{--}10 \text{ MW m}^{-2}$; the ash is pumped from the divertor plenum in that same geometry. Protecting the bred fuel in the exhaust stream is

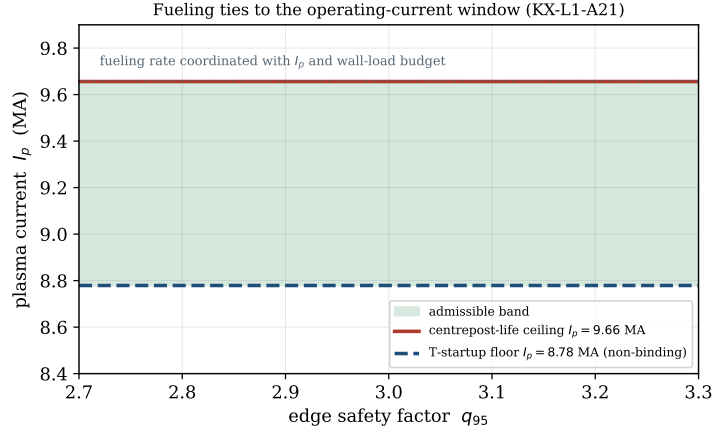


Figure 9. The fueling rate is coordinated with the operating-current window (KX-L1-A21): $I_p \in [8.78, 9.66]$ MA over $q_{95} \in [2.7, 3.3]$, upper-bounded by the centrepost-life ceiling and lower-bounded by a (non-binding) tritium-startup floor. The density controller runs inside this band, not open-loop.

a chemistry problem as much as a pumping one: a flowing-lithium plasma-facing surface retains tritium far better than tin (BR-SX-03/HX-21 rank liquid-Li permeation roughly 6.7 orders of magnitude below Sn, with permeation-barrier coatings ranked $\text{Al}_2\text{O}_3 > \text{Er}_2\text{O}_3 > \text{TiC} > \text{TiN}$) [1], so the exhaust stream loses little of the bred fuel it carries. Figure 10 summarises the full source-to-exhaust chain, and figure 11 the deposition geometry.

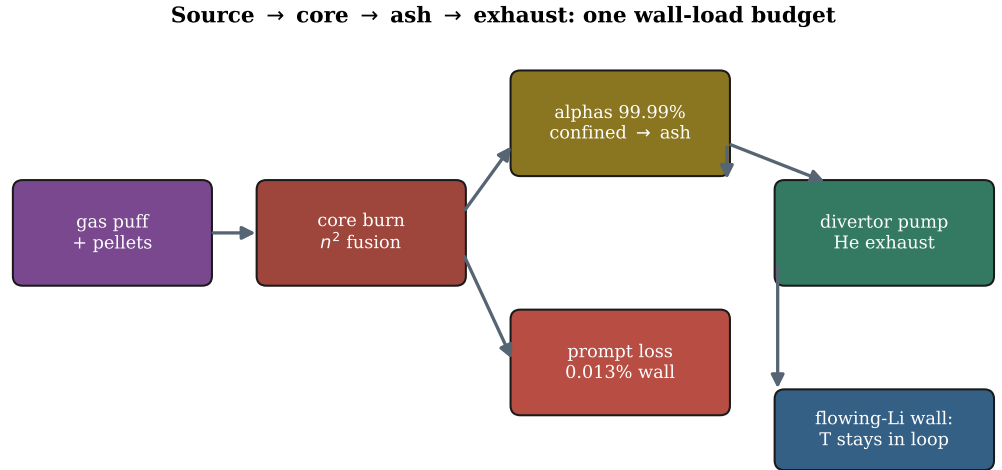


Figure 10. (Schematic.) Source-to-exhaust particle flow: gas and pellets fuel the core; 99.99%-confined alphas thermalize to ash; the divertor pumps helium at the production rate while a flowing-lithium wall keeps the bred tritium in the loop. One wall-load budget governs every step; no data values are shown.

7 Discussion

7.1 Helium exhaust against the published benchmarks

The exhaust specification derived here— $\tau_{\text{He}}^* \lesssim 8.7\text{s}$ for a 5% ash fraction—should be read against the dedicated helium-exhaust experiments that established the achievable range. Reiter *et al.* set the theoretical frame, expressing the burn condition through the ratio $\tau_{\text{He}}^*/\tau_E$ and the divertor enrichment factor of equation (9) [14, 15]. Hillis *et al.* on the TEXTOR ALT-II pump limiter and Wade *et al.* on the DIII-D argon-frosted divertor cryopump both measured helium removal at $\tau_{\text{He}}^* \simeq (7\text{--}20)\tau_E$ [16, 17], and the ITER Physics Basis codified $\tau_{\text{He}}^*/\tau_E \lesssim 10$ as the exhaust target for a burning plasma [18]. For the Hyperion operating point with an energy-confinement time of order $\tau_E \sim 0.3\text{s}$, our 8.7s ceiling is $\tau_{\text{He}}^*/\tau_E \sim 30$, i.e. a *looser* requirement than the experiments have already met—the compact core’s high density and modest ash-dilution tolerance leave more headroom than a large DEMO-class device. The exhaust is therefore not a novel physics gate but

Deposition geometry: deep pellets peak the core

$$\delta = -0.30, \kappa = 2.0, a = 0.48 \text{ m}$$

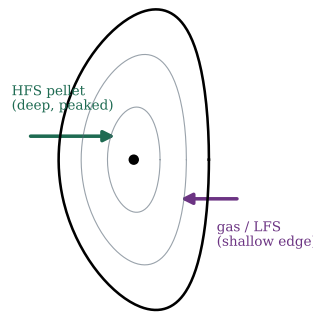


Figure 11. (Schematic.) Deposition geometry in the negative-triangularity ($\delta = -0.30$, $\kappa = 2.0$, $a = 0.48 \text{ m}$) cross-section. Gas and low-field-side sources deposit at the shallow edge front; high-field-side pellets penetrate deep and peak the core, realising the gain of equation (14).

an engineering sizing of divertor conductance in the SOLPS-ITER/EIRENE geometry (§5), and the enrichment relation (9) shows how a mildly enriching divertor keeps τ_{He}^* well below the ceiling.

7.2 Pellet fueling against experiment

The peaking result rests on the well-established experimental fact that HFS pellet launch deposits mass far deeper than LFS launch. Baylor *et al.* on DIII-D produced density-profile peaking factors in excess of three from HFS pellets injected at a factor-of-four lower speed than the LFS reference [12], and Lang *et al.* demonstrated stable high-density operation via pellet fueling on an all-metal wall [13]. Our contribution is not a new fueling technique but the quantitative statement, equation (14), of what peaking is worth *under the compact breeder's neutron-load ceiling*: because average density is pinned by the wall budget rather than by the gas limit, the entire value of pellets is expressed through the $p^2/(2p-1)$ gain, and a demonstrated $p \approx 3$ maps to an 80% fusion-rate increase at fixed wall load. The NGS penetration scaling of equation (16) and figure 7 places the required deposition depth within reach of a few-millimetre HFS pellet, consistent with the injector technology reviewed by Milora and Pégourié [10, 11].

7.3 The density limit in a compact ST

The finding that the neutron wall load, not the Greenwald fraction, sets the ceiling is specific to a nuclear compact ST and inverts the usual DEMO logic. In a conventional device the Greenwald limit [7] is the binding density constraint; here the very high current density pushes n_{GW} to $1.33 \times 10^{21} \text{ m}^{-3}$ (equation 12), so the machine reaches its neutron-load ceiling at $f_G \approx 0.36$ long before the gas limit at $f_G \approx 0.45$ becomes relevant. This is a comfortable margin against the disruptive density limit and is consistent with the ELM-free, quiescent negative-triangularity edge [6, 5] that the machine runs, which removes the pedestal-driven ELM cycle that would otherwise complicate high-density fueling.

7.4 Relation to the Kronos series

This particle balance sits at the centre of the Kronos breeder physics chain. The alpha source and confinement come from the fast-ion study [doi:10.5281/zenodo.22645729; alpha-fast-product-confinement]; the density ceiling connects to the Greenwald-limit and radiative-boundary analysis [doi:10.5281/zenodo.22645947; greenwald-fraction-density-limit]; the ash-exhaust path is set by the stacked detachment portfolio [doi:10.5281/zenodo.22645742; divertor-portfolio] and the blanket/divertor thermal-hydraulics [doi:10.5281/zenodo.22645836; blanket-and-divertor-thermal-hydraulics]; the wall-chemistry choice that keeps bred tritium in the loop is the permeation-barrier-coating study [doi:10.5281/zenodo.22645870; tritium-permeation-barrier-coatings]. Downstream, the surplus tritium the breeder banks decays to the helium-3 that fuels the D-³He burner, whose fuel-cycle closure is treated in the tritium-breeding ladder [doi:10.5281/zenodo.22645746; tritium-fuel-cycle-and-breeding-ladder] and the helium-3 supply analysis [doi:10.5281/zenodo.22645734; helium-3-fuel-cycle]; the fueling

actuator itself is one controller in the breeder real-time control suite [doi:10.5281/zenodo.22645809; real-time-control-suite].

8 Limitations

The pellet-peaking benefit is quantified here as a scaling argument: equation (14) gives the fusion-rate gain for a given peaking factor p , and the NGS scaling of equation (16) estimates the penetration depth of a given pellet, but the step that closes the loop—converting a deposited profile into the achieved steady-state p and the associated confinement time—requires a self-consistent core particle-transport calculation with the anomalous pinch [19] resolved on the negative-triangularity turbulence. BR-CX-09 flags exactly this as requiring a dedicated transport code [1], and the one-dimensional solver of §4 is built to consume the turbulent D_s, v_s from that code once run. Until then, the peaking gain is an upper-bound scaling, not a predicted operating value; the source-to-exhaust budget of §2 and the exhaust and dilution relations of §3, which do not depend on the peaking closure, stand on the frozen anchors.

9 Conclusion

The Hyperion breeder’s particle balance closes from source to exhaust at the frozen design point. The core burns D–T at $6.0 \times 10^{19} \text{ s}^{-1}$ and makes helium ash at $3.0 \times 10^{19} \text{ s}^{-1}$ with 99.99% alpha confinement (table 1); the operating density is capped at $f_G \approx 0.36$ ($\bar{n} = 4.8 \times 10^{20} \text{ m}^{-3}$) by the first-wall neutron budget rather than by the $f_G \approx 0.45$ gas limit (figure 2); pellet injection adds core peaking within that ceiling, worth $g(p) = p^2/(2p - 1)$ in fusion rate— $\times 1.33$ at $p = 2$, $\times 1.80$ at $p = 3$ (figure 8)—at fixed wall load; and the divertor must exhaust helium at the production rate, holding the effective ash confinement time below $\tau_{\text{He}}^* \lesssim 8.7 \text{ s}$ for a 5% ash fraction (figure 3), a requirement looser than dedicated exhaust experiments have already met. A flowing-lithium plasma-facing surface keeps the bred tritium in the loop through the exhaust, and the fueling rate is coordinated with the operating-current window rather than run open-loop. Fueling method, density ceiling, and ash exhaust are one coupled system, governed by the single wall-load constraint that also governs the rest of the machine.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The authors thank the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim.

Reproducibility. The particle-balance rates are derived from the frozen anchor $P_{\text{fus}} = 85.04 \text{ MW}$ through equations (1)–(2); the density-ceiling, alpha-confinement, ash and pellet quantities are frozen results (BR-CX-09, BR-L1-A14, BR-L2-A20, BR-L1-A6, KX-L1-A21, BR-SX-15, BR-SX-03) in the Kronos Physics De-Risking Register [1], doi:10.5281/zenodo.22133057. Every figure is regenerated byte-for-byte by the deposited script `make_fueling_ash_figs_2p45.py`. This paper is archived at its DOI doi:10.5281/zenodo.22645848; official version: <https://www.kronosfusionenergy.com/publications/pellet-fueling-particle-balance-and-helium-ash-exhaust.html> and its official page is [kronosfusionenergy.com/publications/pellet-fueling-particle-balance-and-helium-ash-exhaust](https://www.kronosfusionenergy.com/publications/pellet-fueling-particle-balance-and-helium-ash-exhaust).

Data availability

The balance inputs and figure scripts are archived with the deposit at doi:10.5281/zenodo.22645848 and reference the Kronos 2026 series and its Physics De-Risking Register (doi:10.5281/zenodo.22133057). All quantitative values trace to the register by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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